_eda_price_2025

March 30, 2026

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import subprocess
import os
from scipy.stats import chi2_contingency
from datetime import timedelta

from etl.customer import aggregate_customer
from etl.subsidy import report_customer_conversion_by_subsidy_issue_status,␣
↪report_customer_metrics_by_subsidy_issue_status
from etl.subsidy import visualize_conversion_by_subsidy_issues,␣
↪report_customer_subsidy_issues_by_product
from etl.subsidy import visualize_customer_subsidy_issues_by_product,␣
↪visualize_policy_suspension_impact
from etl.price import visualize_conversion_by_price_sweet_spots,␣
↪visualize_conversion_by_price_sweet_spots_variant
from etl.price import report_product_conversion_by_price_quartile,␣
↪visualize_conversion_by_equipment_category_price
from etl.price import visualise_conversion_by_brand_price,␣
↪show_two_panel_conversion_by_brand_price
from etl.price import visualize_conversion_by_equipment_type_brand_price,␣
↪show_conversion_by_price_dashboard,␣
↪show_product_type_price_conversion_heatmap
from etl.season import visualize_conversion_by_season,␣
↪visualize_regional_seasonality
from etl.period import visualize_conversion_by_year,␣
↪report_best_and_worst_months, show_conversion_by_price_over_time,␣
↪overlay_all_years_on_one_plot
from etl.period import visualize_conversion_by_prices_sweet_spot_shift,␣
↪visualize_decision_times, show_decision_time_dashboard
from etl.product import visualise_heat_pump_performance,␣
↪show_product_type_by_brand_heatmap, show_product_type_by_agency_heatmap,␣
↪show_brand_by_product_type_heatmap
```

```python
import warnings
warnings.filterwarnings('ignore')

df_clean = pd.read_csv('cleaned_quote_data.csv')
df_clean['dt_creation_devis'] = pd.to_datetime(df_clean['dt_creation_devis'])
print(f"Loaded data: {len(df_clean):,} quotes from {df_clean['numero_compte'].
 ↪nunique():,} customers")
```

```
Loaded data: 37,008 quotes from 25,716 customers
```

```python
#customers = aggregate_customer(df_clean)
```

```python
all_customers = pd.read_csv('customer_master_data.csv')
```

```python
# Method 1: Using year column
customers_2025 = all_customers[all_customers['year'] == 2025]

# Method 2: Using period column
customers_2025 = all_customers[all_customers['period'] == '2025']

# Method 3: Using date range
customers_2025 = all_customers[
    (all_customers['first_quote_date'] >= '2025-01-01') &
    (all_customers['first_quote_date'] <= '2025-12-31')
]

print(f"2025 customers: {len(customers_2025):,}")
```

```
2025 customers: 9,944
```

```python
customers = customers_2025
```

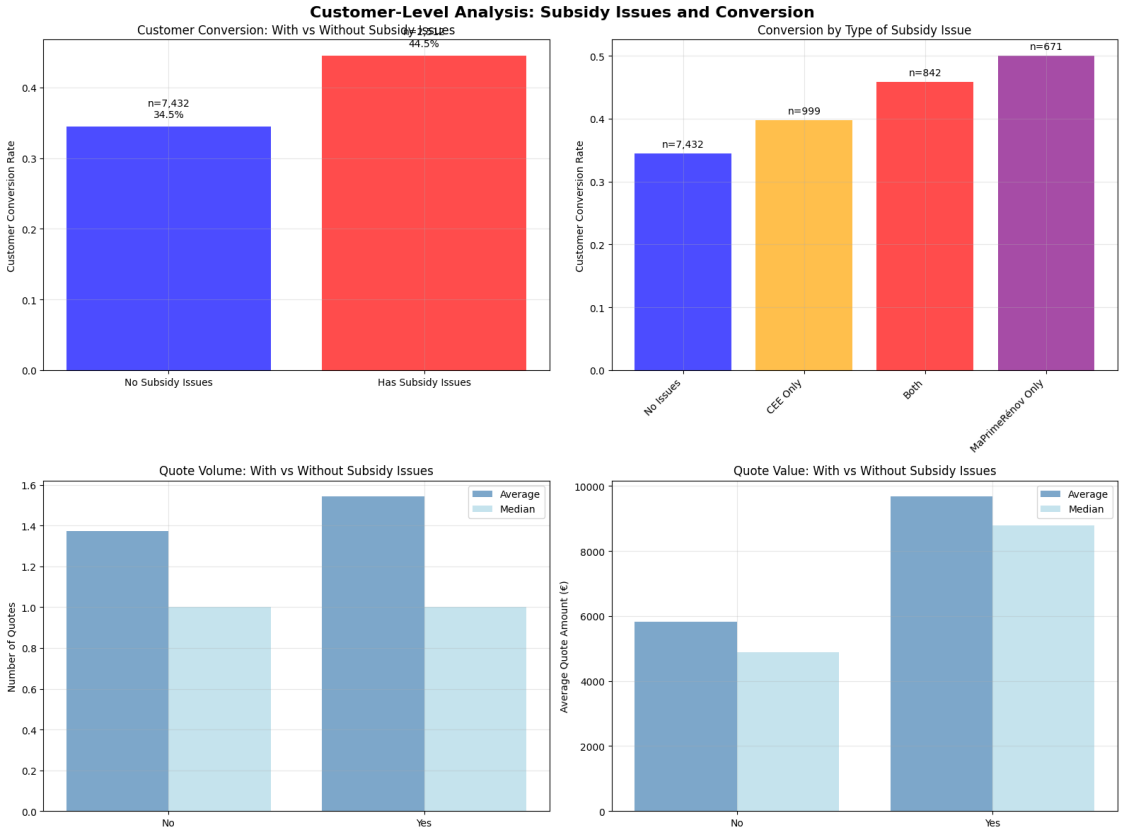
```python
conv_no_issue, conv_with_issue =␣
 ↪report_customer_conversion_by_subsidy_issue_status(customers)
report_customer_metrics_by_subsidy_issue_status(customers)
visualize_conversion_by_subsidy_issues(customers, conv_no_issue,␣
 ↪conv_with_issue)
visualize_policy_suspension_impact(customers, price_var='max_out_of_pocket')

top_products, no_issue_pct, issue_pct =␣
 ↪report_customer_subsidy_issues_by_product(customers)
visualize_customer_subsidy_issues_by_product(customers, top_products,␣
 ↪no_issue_pct, issue_pct)

visualize_conversion_by_price_sweet_spots(customers,␣
 ↪price_var='max_out_of_pocket')
```

```
visualize_conversion_by_price_sweet_spots_variant(customers,␣
 ↪price_var='max_out_of_pocket')
report_product_conversion_by_price_quartile(customers)
visualize_conversion_by_equipment_category_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualise_conversion_by_brand_price(customers, price_var='max_out_of_pocket')
show_two_panel_conversion_by_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_equipment_type_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')

visualize_conversion_by_season(customers, price_var='max_out_of_pocket')
visualize_regional_seasonality(customers, price_var='max_out_of_pocket')

visualize_conversion_by_year(customers, price_var='max_out_of_pocket')
report_best_and_worst_months(customers, price_var='max_out_of_pocket')
show_conversion_by_price_over_time(customers, price_var='max_out_of_pocket')
overlay_all_years_on_one_plot(customers, price_var='max_out_of_pocket')
visualize_conversion_by_prices_sweet_spot_shift(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_decision_times(customers, price_var='max_out_of_pocket')
show_product_type_price_conversion_heatmap(customers,␣
 ↪price_var='max_out_of_pocket')
show_decision_time_dashboard(customers, price_var='max_out_of_pocket')

visualise_heat_pump_performance(customers, price_var='max_out_of_pocket')

show_conversion_by_price_dashboard(customers, price_var='max_out_of_pocket')

show_product_type_by_brand_heatmap(customers)
show_product_type_by_agency_heatmap(customers)
show_brand_by_product_type_heatmap(customers)
```

```
Customer conversion by subsidy issue status:

                   conversion_rate  customer_count  converted_count
had_subsidy_issue
False                     0.344591            7432           2561.0
True                      0.445462            2512           1119.0

Chi-square p-value: 1.741976e-19
Statistically significant: YES

Conversion without issues: 34.46%
Conversion with issues: 44.55%
Difference: 10.1 percentage points
```

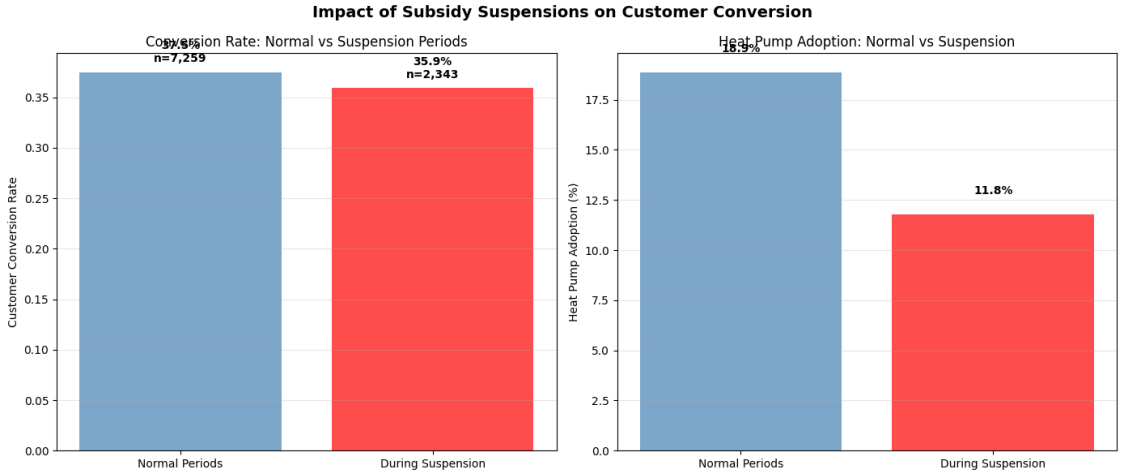
```
Customer-level finding: +10.1 points

================================================================================
Customer characteristics by subsidy issue status
================================================================================

Metric comparison:

total_quotes:
  No issues: 1.37
  With issues: 1.54
  Ratio: 1.12x

avg_quote_amount:
  No issues: 5824.46
  With issues: 9681.25
  Ratio: 1.66x

max_quote_amount:
  No issues: 6196.97
  With issues: 10553.75
  Ratio: 1.70x

customer_duration_days:
  No issues: 6.85
  With issues: 9.40
  Ratio: 1.37x

price_range:
  No issues: 819.22
  With issues: 1660.17
  Ratio: 2.03x

multiple_quotes:
  No issues: 0.28
  With issues: 0.37
  Ratio: 1.32x

Quote pattern distribution by subsidy issue:
had_subsidy_issue  False  True
quote_pattern
Single quote       0.720   0.63
Three+ quotes      0.064   0.11
Two quotes         0.216   0.26

Decision speed by subsidy issue:
had_subsidy_issue   False  True
decision_speed
```

```
Long (>30 days)     0.046  0.074
Medium (8-30 days)  0.036  0.051
Quick (2-7 days)    0.044  0.075
Same day decision   0.874  0.800

Product family by subsidy issue status:
had_subsidy_issue                 False  True
main_product_family
Appareil hybride                  0.004  0.037
Autres                            0.009  0.004
Chaudière                         0.391  0.063
Climatisation                     0.294  0.100
ECS : Chauffe-eau ou adoucisseur  0.057  0.020
Emetteur de chauffage  ou chappe  0.005  0.004
Fumisterie                        0.001  0.000
Photovoltaïque                    0.025  0.011
Plomberie Sanitaire               0.008  0.000
Pompe à chaleur                   0.050  0.395
Poêle                             0.151  0.361
Produit VMC                       0.005  0.000
Unknown                           0.001  0.003

Conversion by subsidy issue type:
                        mean  count
subsidy_issue_type
MaPrimeRénov Only   0.500745    671
Both                0.458432    842
CEE Only            0.397397    999
No Issues           0.344591   7432
```



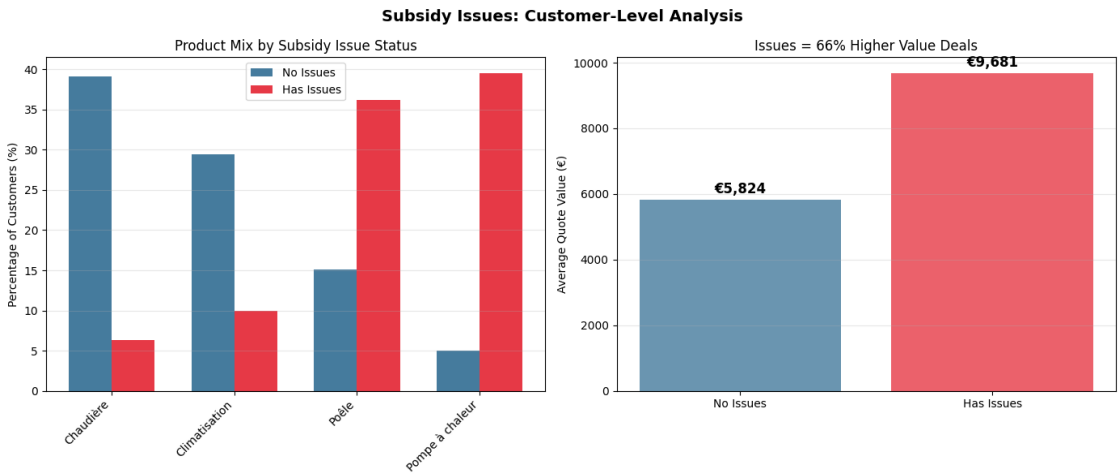

```
================================================================================
Policy Suspension Impact
================================================================================

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191
Unique values in during_suspension: [False  True]

Conversion during subsidy suspensions:
                       mean  count
Normal Periods     0.374845   7259
During Suspension  0.358942   2343

Suspension impact p-value: 0.1735
Statistically significant: NO

Product mix during suspensions:
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
during_suspension
```

False	19.5	33.4	16.4	8.6	21.9	0.2
True	44.3	24.6	9.4	6.2	15.5	0.0

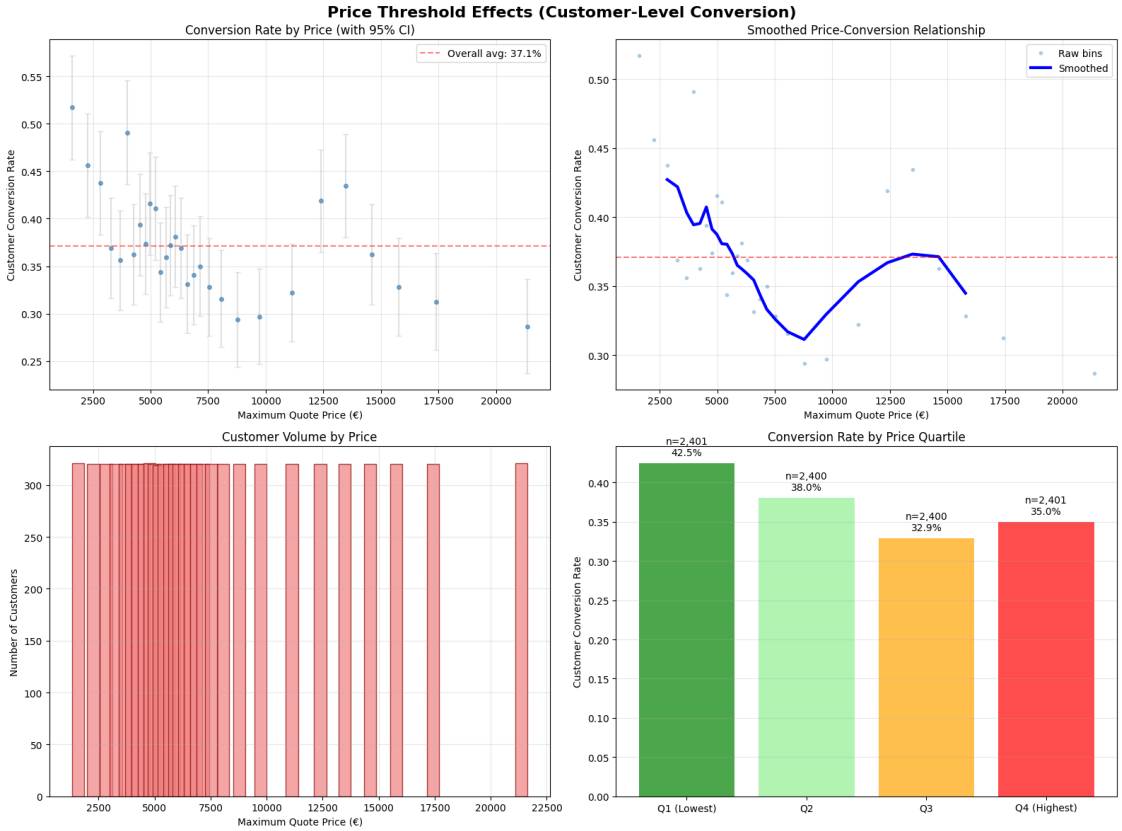



```
Unique values in main_product_family:
main_product_family
Chaudière                          3065
Climatisation                      2438
Poêle                              2027
Pompe à chaleur                    1362
ECS : Chauffe-eau ou adoucisseur    471
Photovoltaïque                      216
Appareil hybride                    122
Autres                               79
Plomberie Sanitaire                  57
Emetteur de chauffage  ou chappe     47
Produit VMC                          37
Unknown                              16
Fumisterie                            7
Name: count, dtype: int64

Top products:
main_product_family
Chaudière                          3065
Climatisation                      2438
Poêle                              2027
Pompe à chaleur                    1362
ECS : Chauffe-eau ou adoucisseur    471
Photovoltaïque                      216
Appareil hybride                    122
Autres                               79
Plomberie Sanitaire                  57
```

```
Emetteur de chauffage  ou chappe      47
Name: count, dtype: int64

Using these products: ['Chaudière', 'Climatisation', 'Poêle', 'Pompe à chaleur']

No Issue %: [np.float64(39.10118406889128), np.float64(29.440258342303554),
np.float64(15.056512378902045), np.float64(4.965016146393972)]
Has Issue %: [np.float64(6.329617834394904), np.float64(9.952229299363058),
np.float64(36.146496815286625), np.float64(39.53025477707006)]
```



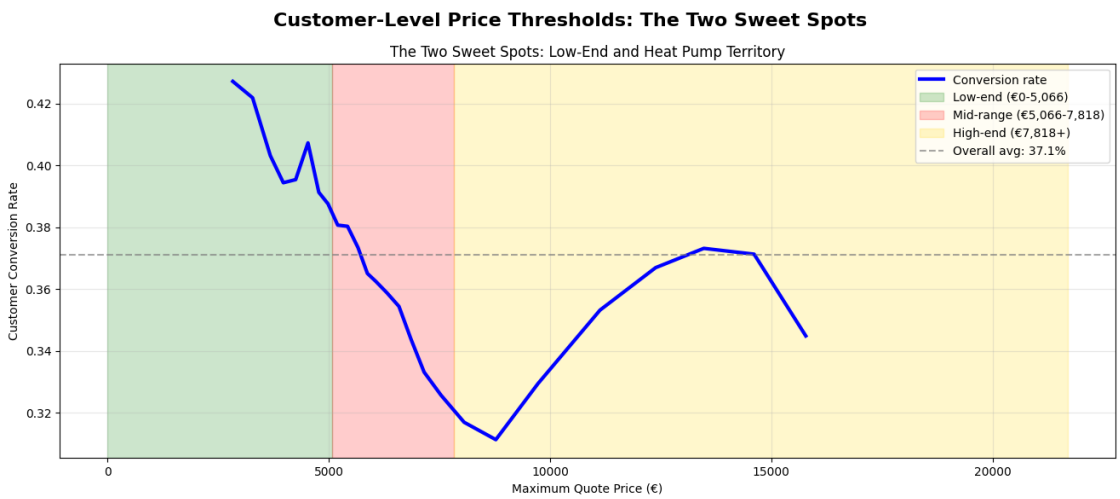

```
================================================================================
Threshold effect
================================================================================

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191
```

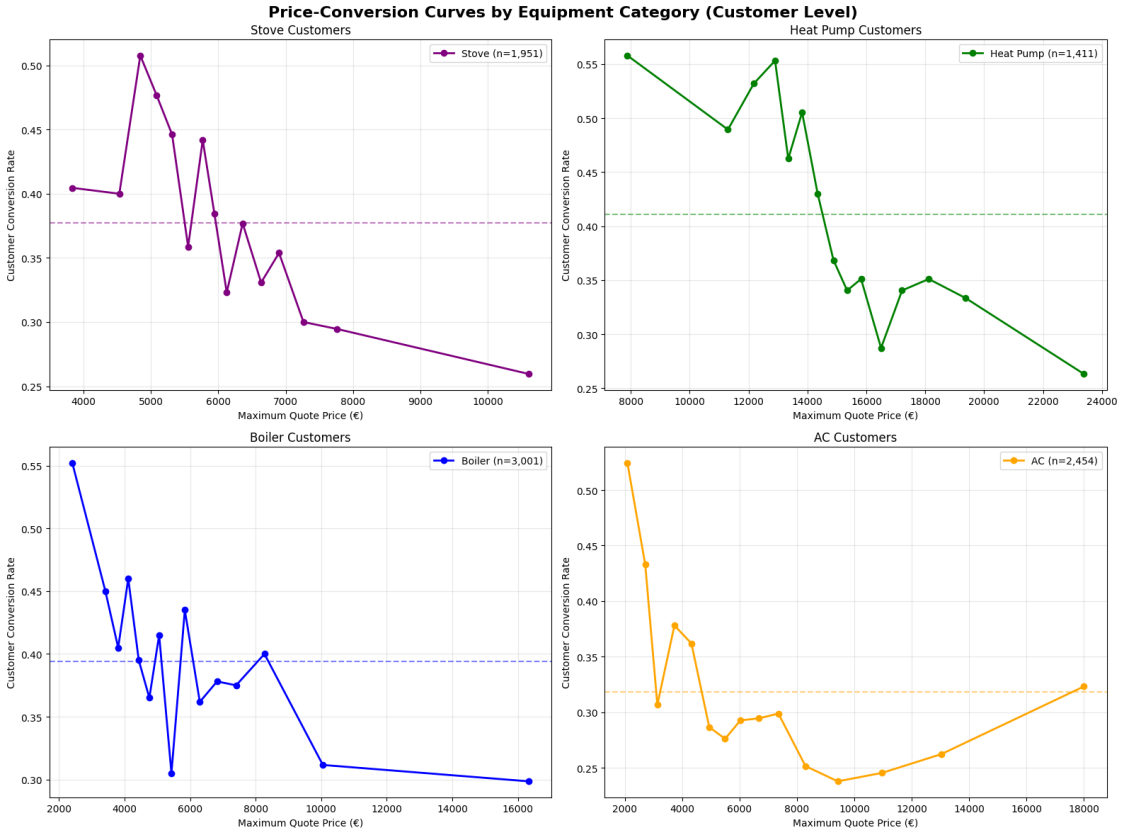



```
================================================================================
THRESHOLD EFFECT: KEY FINDINGS
================================================================================

Conversion by Price Quartile:
  Q1 (Lowest): 42.5% (n=2,401)
  Q2: 38.0% (n=2,400)
  Q3: 32.9% (n=2,400)
  Q4 (Highest): 35.0% (n=2,401)

Difference between lowest and highest quartile: 7.5 percentage points

Sweet spots (above-average conversion):
  €1,595: 51.7% (n=321)
  €2,261: 45.6% (n=320)
  €2,830: 43.8% (n=320)
  €3,972: 49.1% (n=320)
  €4,529: 39.4% (n=320)
  €4,774: 37.4% (n=321)
  €4,982: 41.6% (n=320)
```

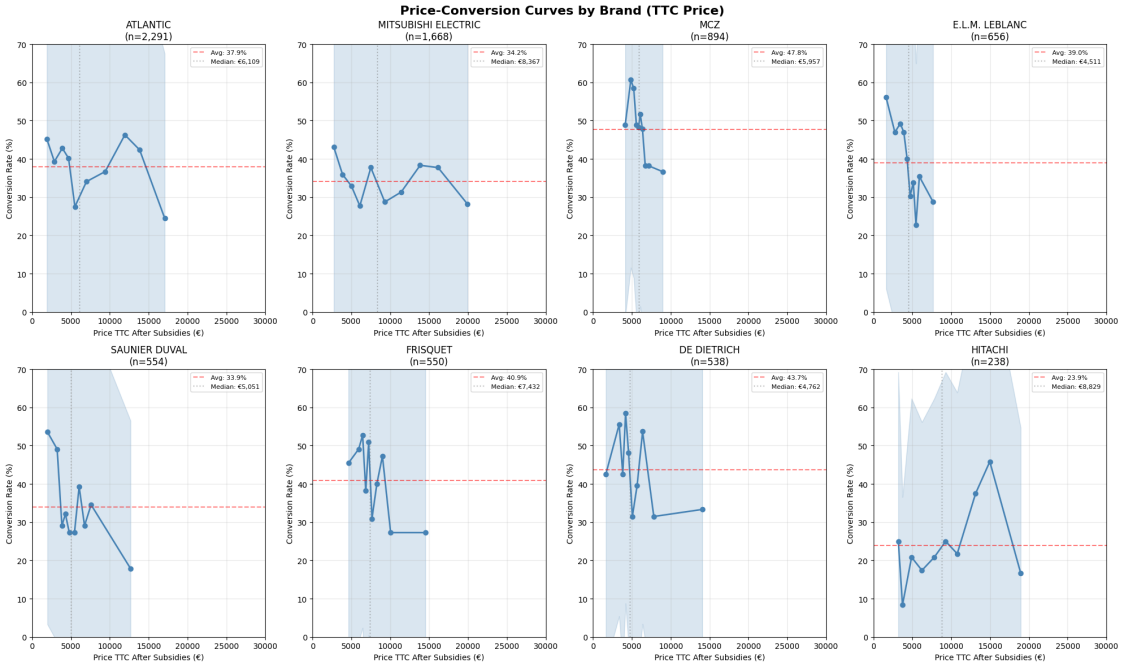
```
  €5,201: 41.1% (n=319)
  €5,872: 37.2% (n=320)
  €6,059: 38.1% (n=320)
  €12,387: 41.9% (n=320)
  €13,476: 43.4% (n=320)

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191

================================================================================
Product mix by price quartile
================================================================================

Product distribution by price quartile (%):
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
price_quartile
Q1_Low                   31.9    38.8        0.6   21.3    7.3      0.2
Q2_MidLow                20.1    37.7        0.6    2.3   39.2      0.1
Q3_MidHigh               27.4    34.7        1.9    4.7   31.2      0.1
Q4_High                  22.0    12.5       56.6    6.5    2.1      0.3

================================================================================
Conversion rates by product and price quartile
================================================================================

Heat Pump (n=1,525):
  Q1_Low: 21.4%
  Q2_MidLow: 64.3%
  Q3_MidHigh: 57.4%
  Q4_High: 39.7%

Boiler (n=3,032):
  Q1_Low: 45.1%
  Q2_MidLow: 38.6%
  Q3_MidHigh: 37.2%
  Q4_High: 30.4%

AC (n=2,491):
  Q1_Low: 39.9%
  Q2_MidLow: 29.9%
  Q3_MidHigh: 26.9%
  Q4_High: 27.8%

Stove (n=2,024):
  Q1_Low: 41.9%
  Q2_MidLow: 41.8%
  Q3_MidHigh: 32.3%
```

```
  Q4_High: 30.8%

================================================================================
HEAT PUMP CUSTOMERS (n=1,774)
================================================================================
Average price: €15,786
Conversion rate: 39.5%
Subsidy issue rate: 73.3%

Heat pump adoption by quartile:
  Q1_Low: 0.7%
  Q2_MidLow: 0.7%
  Q3_MidHigh: 2.5%
  Q4_High: 65.9%

================================================================================
Equipment Category Segment Effect
================================================================================

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191
```

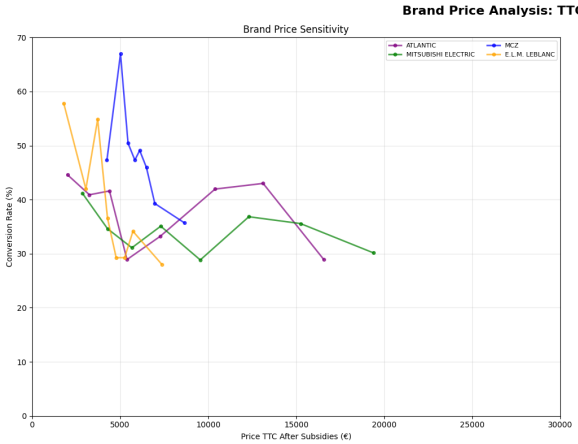





```
================================================================================
Conversion by Brand Price
================================================================================

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191
Top 8 brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'MCZ', 'E.L.M. LEBLANC',
'SAUNIER DUVAL', 'FRISQUET', 'DE DIETRICH', 'HITACHI']
```




```
Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191
```

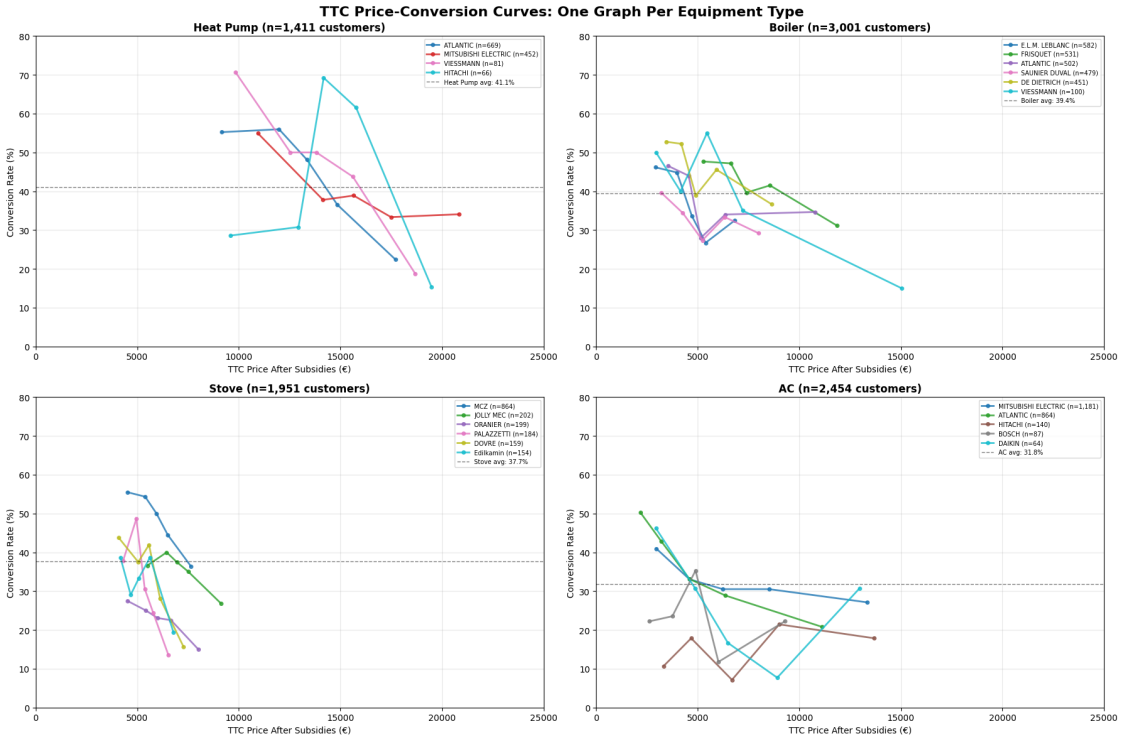


Brand Performance Summary

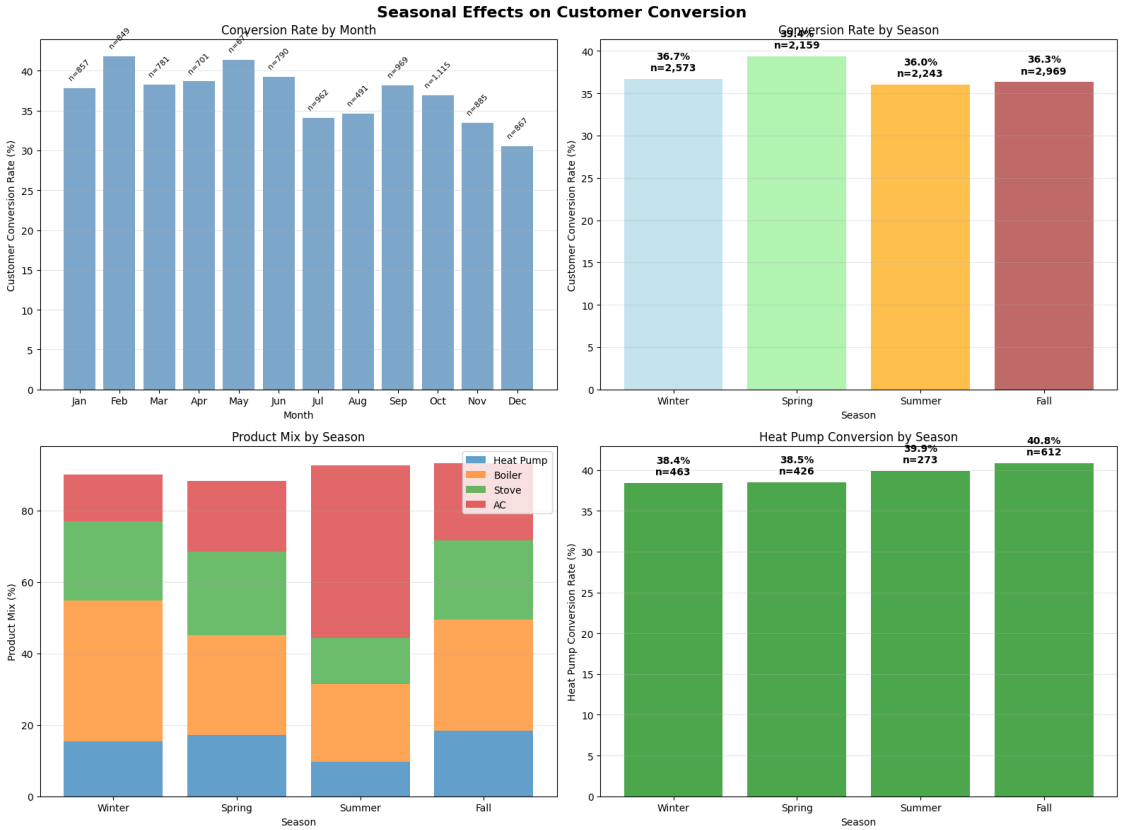
Brand	Customers	Avg Conv	Avg Price	Best Price Range	Best Conv
ATLANTIC	2,291	37.9%	€7,801	€1,338-3,400	42.3%
MITSUBISHI ELECTRIC	1,668	34.2%	€9,580	€1,368-4,405	39.5%
MCZ	894	47.8%	€6,098	€1,800-5,067	54.7%
E.L.M. LEBLANC	656	39.0%	€4,495	€1,347-3,237	51.5%
SAUNIER DUVAL	554	33.9%	€5,660	€1,332-3,580	51.4%
FRISQUET	550	40.9%	€8,052	€2,391-6,228	47.3%
DE DIETRICH	538	43.7%	€5,665	€3,654-4,427	50.5%
HITACHI	238	23.9%	€9,312	€13,879-25,964	31.2%

```
================================================================================
TTC Price-conversion curves: 4 equipment types, multiple brands per graph
================================================================================

Analyzing 9,602 customers
```

```
Price range: €1,332 to €27,121
Median price: €6,191

Heat Pump - Top brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'VIESSMANN',
'HITACHI']

Boiler - Top brands: ['E.L.M. LEBLANC', 'FRISQUET', 'ATLANTIC', 'SAUNIER DUVAL',
'DE DIETRICH', 'VIESSMANN']

Stove - Top brands: ['MCZ', 'JOLLY MEC', 'ORANIER', 'PALAZZETTI', 'DOVRE',
'Edilkamin']

AC - Top brands: ['MITSUBISHI ELECTRIC', 'ATLANTIC', 'HITACHI', 'BOSCH',
'DAIKIN']
```

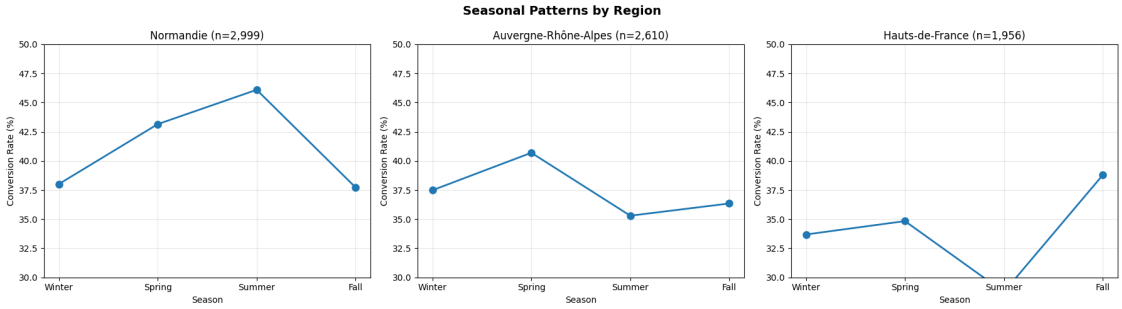



```
================================================================================
Seasonal Effects on Customer Conversion
================================================================================
```

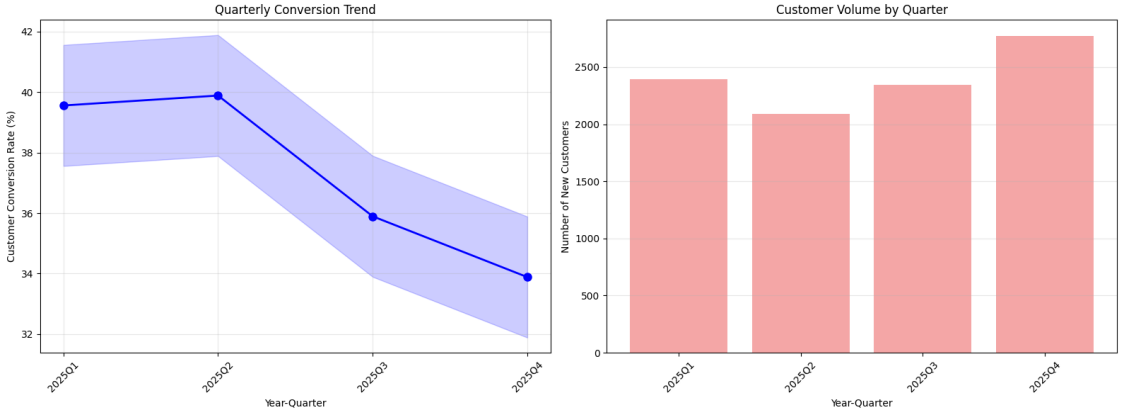



```
================================================================================
SEASONAL STATISTICS
================================================================================

Conversion by season:
  Winter: 36.7% (n=2,573)
  Spring: 39.4% (n=2,159)
  Summer: 36.0% (n=2,243)
  Fall: 36.3% (n=2,969)

Seasonality chi-square p-value: 0.0773
Statistically significant: NO

================================================================================
Regional Seasonal Pattern
================================================================================

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191
```

```
Top regions: ['Normandie', 'Auvergne-Rhône-Alpes', 'Hauts-de-France']
```

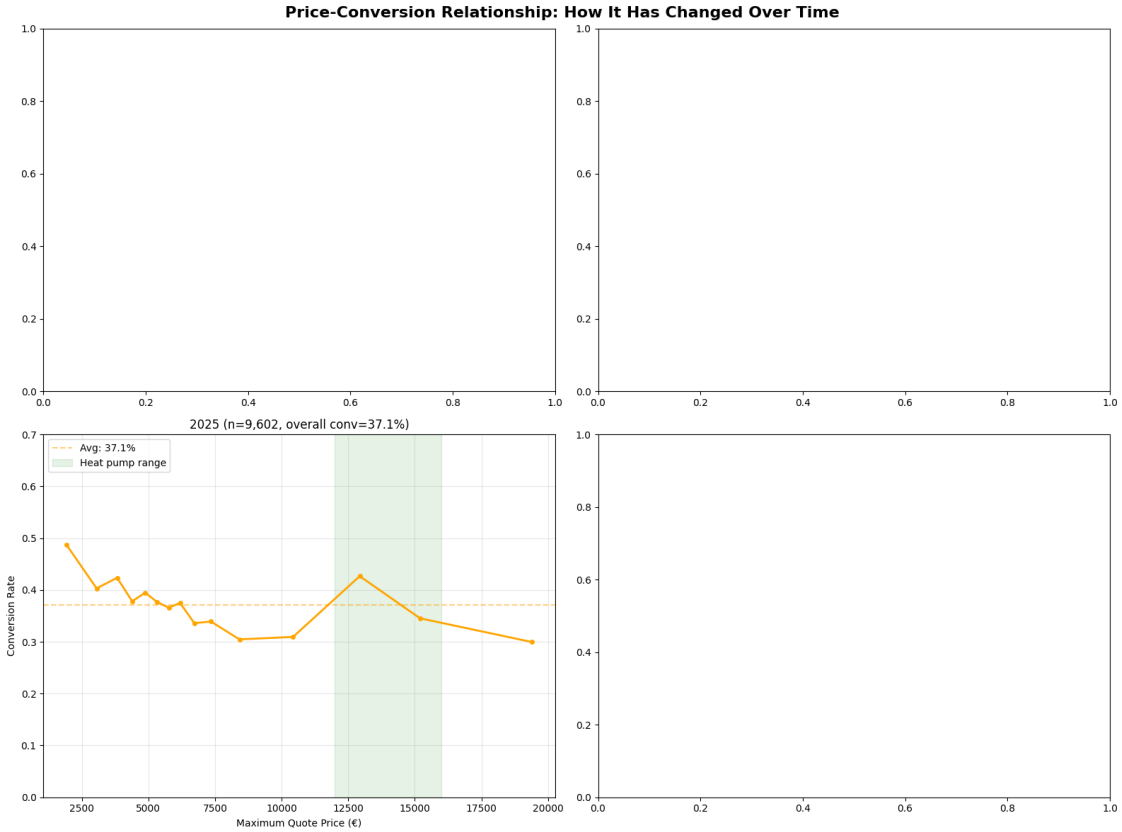



```
================================================================================
Year-over-year trends
================================================================================

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191

Quarterly conversion rates:
                  mean  count
year_quarter
2025Q1        0.395572   2394
2025Q2        0.398852   2091
2025Q3        0.358942   2343
2025Q4        0.338861   2774
```

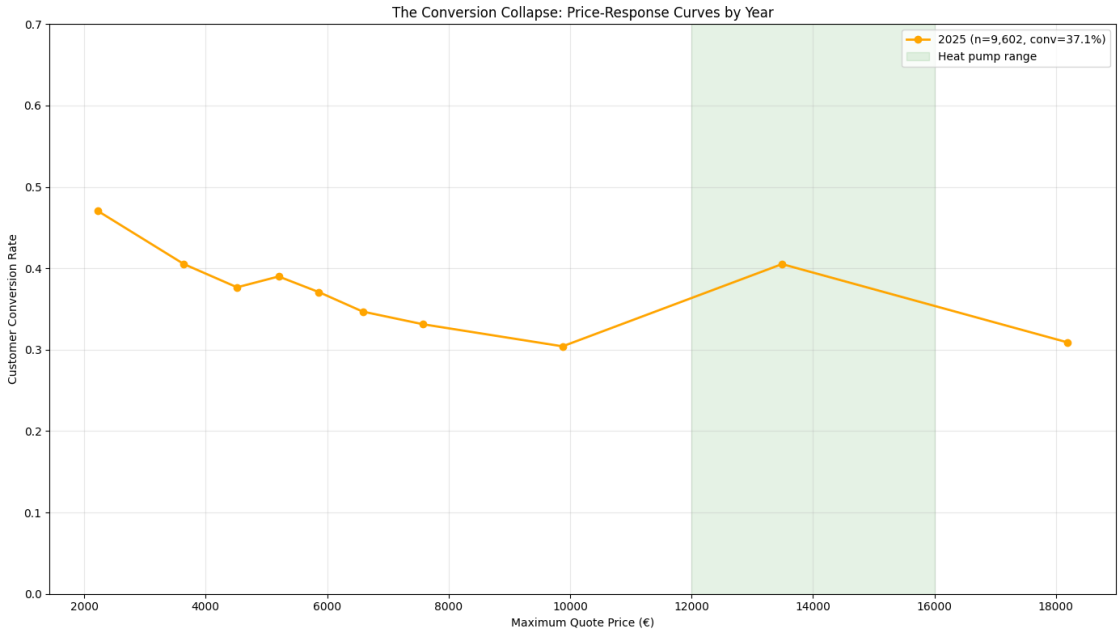



```
================================================================================
```

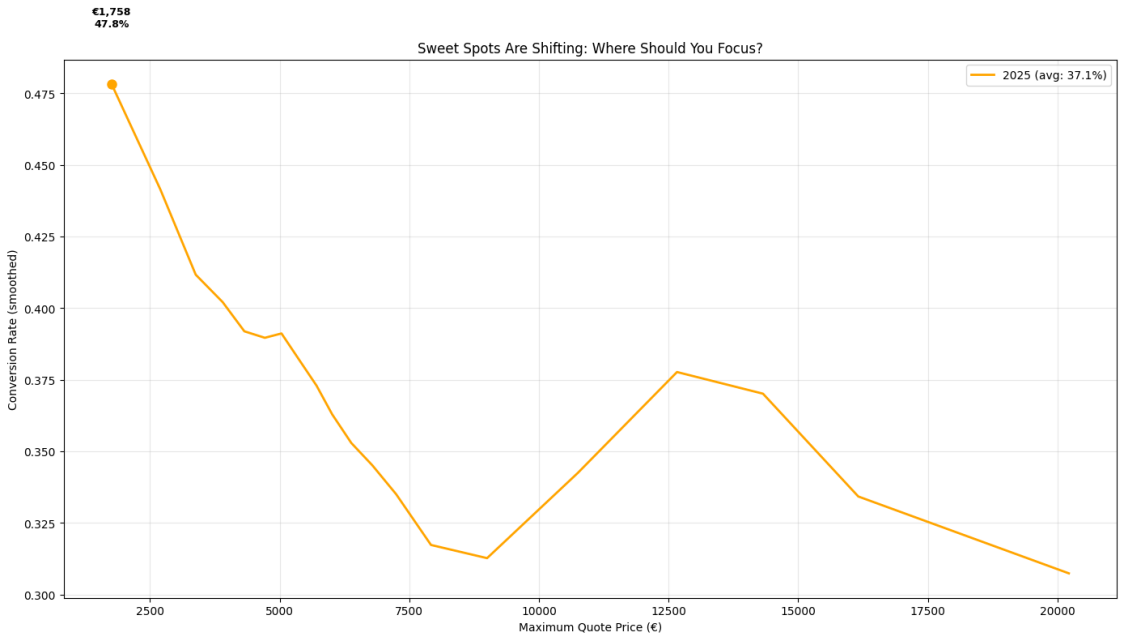
```
Best and Worst Months Report
================================================================================

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191

Conversion during subsidy suspensions:
                       mean  count
Normal Periods     0.374845   7259
During Suspension  0.358942   2343

Suspension impact p-value: 0.1735
Statistically significant: NO

Best month: 2 (42.4%, n=816)
Worst month: 12 (30.8%, n=842)

Best season: Spring (39.5%)
Worst season: Fall (36.1%)

Heat pump conversion by season:
  Winter: 39.6%
  Spring: 39.5%
  Summer: 42.0%
  Fall: 41.4%

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191
```



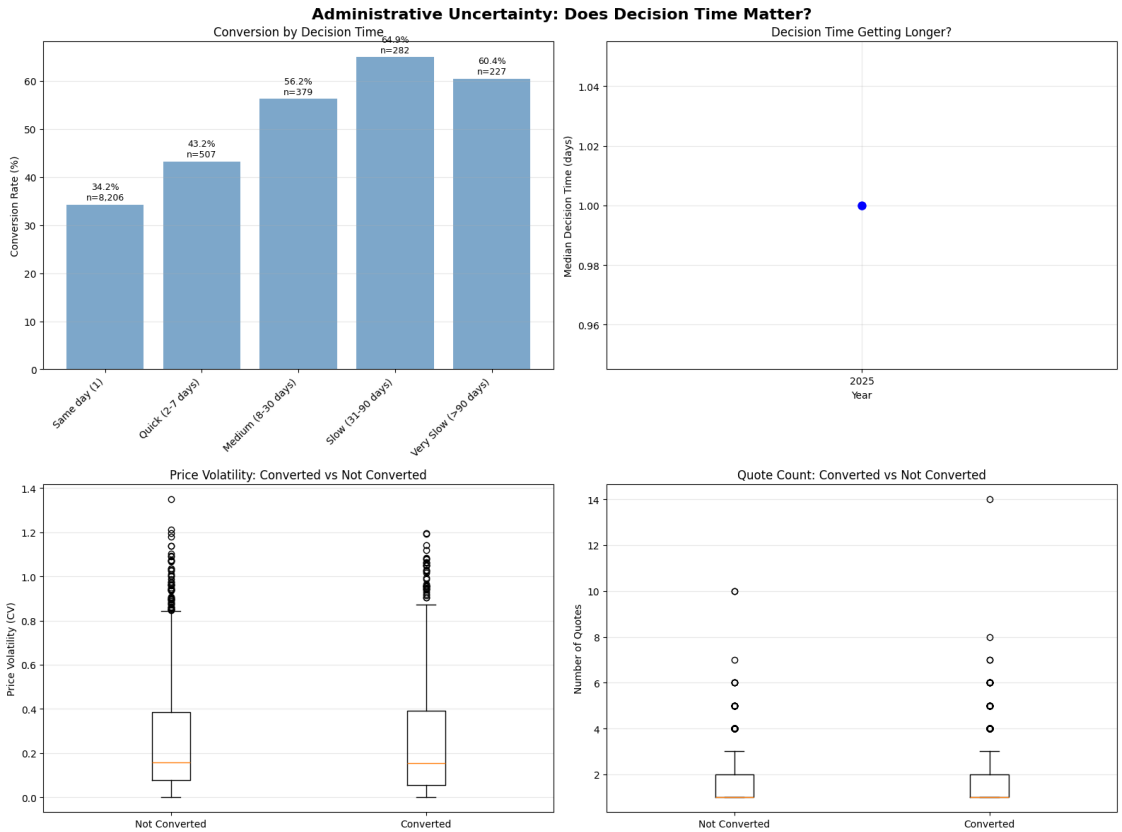

```
Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191
```




```
================================================================================
Sweet Spot Analysis Over Time
================================================================================

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191

Sweet spots by year (price points with conversion > avg + 3%):

2025 (overall avg: 37.1%):
  €1,758: 50.9% (n=481)
  €2,698: 43.1% (n=480)
  €3,902: 43.1% (n=480)
  €5,036: 41.9% (n=480)
  €12,663: 43.3% (n=480)
```

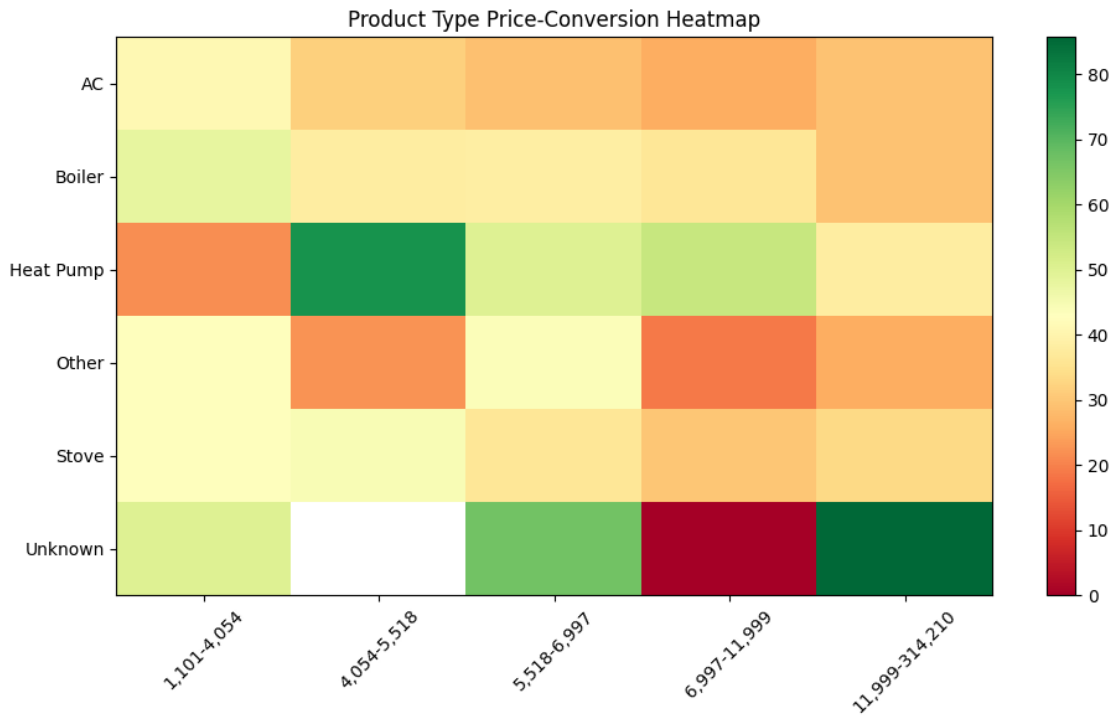



```
================================================================================
Administrative Uncertainty Effect
================================================================================

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191

Decision time distribution:
count    9602.000000
mean        7.459175
std        30.155039
min         1.000000
25%         1.000000
50%         1.000000
75%         1.000000
max       372.000000
Name: decision_days, dtype: float64

Customers with 1-day decision: 8,206 (85.5%)
```



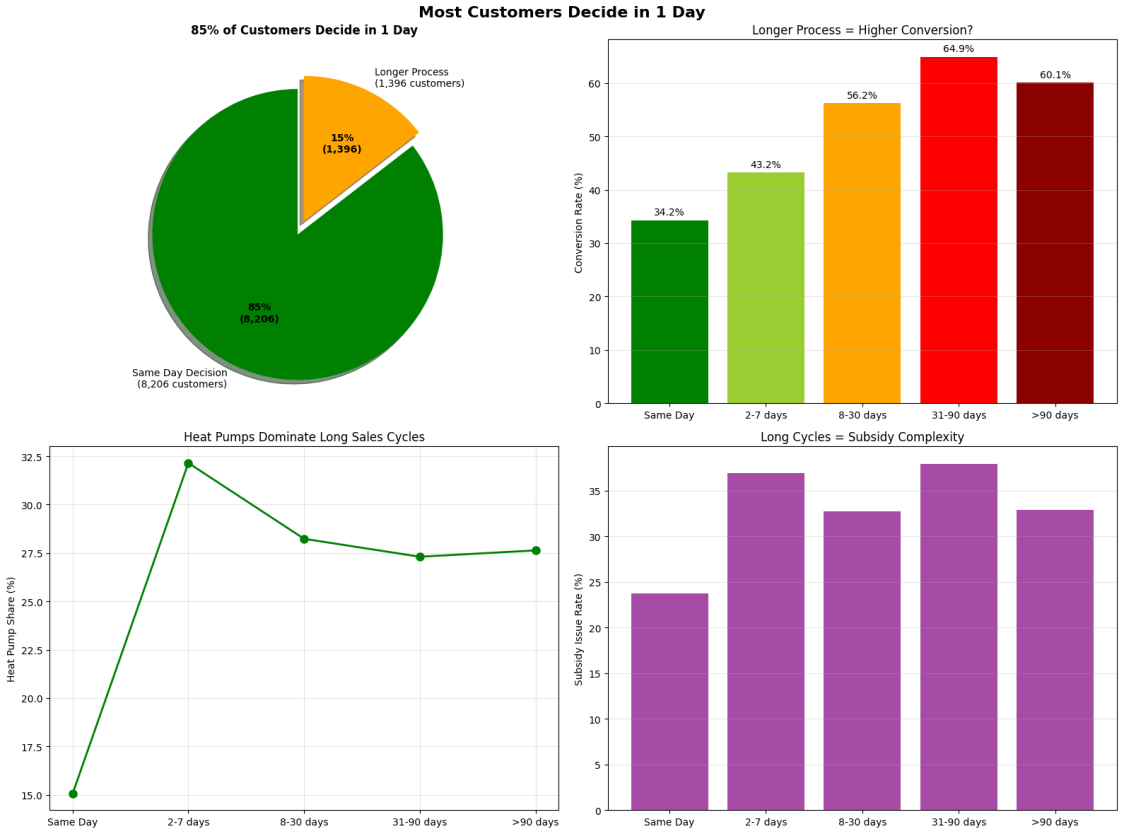

```
================================================================================
Product Type - Price Conversion Heatmap
================================================================================
Simple pivot table:
price_bin_simple         (1101.459, 4054.72]  (4054.72, 5518.8]  \
main_equipment_category
AC                                 40.902256          31.630170
Boiler                             48.093842          38.122172
Heat Pump                          21.428571          77.777778
Other                              42.829077          22.222222
Stove                              43.023256          44.264943
Unknown                            50.000000                NaN

price_bin_simple         (5518.8, 6997.32]  (6997.32, 11999.94]  \
main_equipment_category
AC                                28.850856           25.914634
Boiler                            38.325282           36.305732
Heat Pump                         50.000000           54.347826
Other                             43.589744           19.047619
Stove                             36.244019           29.947917
```

```
Unknown                          66.666667            0.000000

price_bin_simple         (11999.94, 314210.61]
main_equipment_category
AC                                   29.154519
Boiler                               29.439252
Heat Pump                            38.083736
Other                                26.016260
Stove                                33.333333
Unknown                              85.714286
```



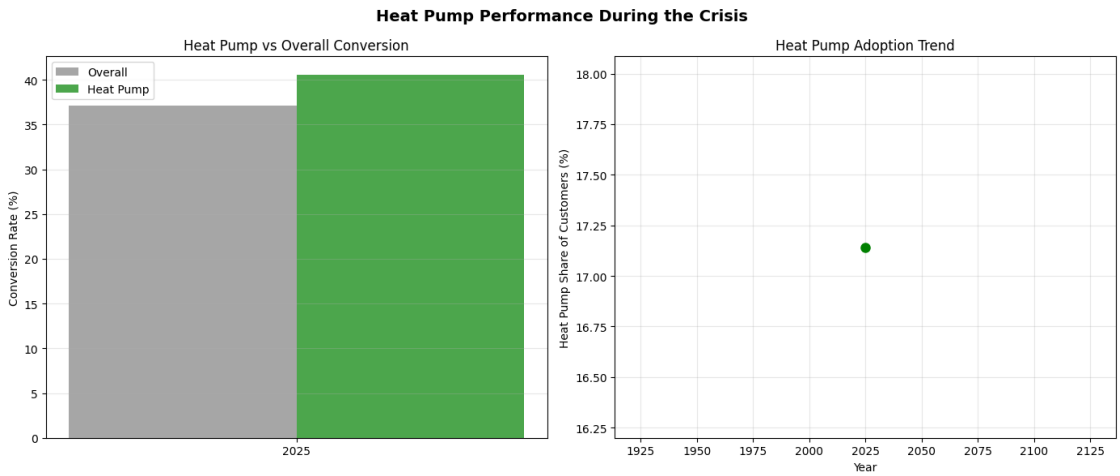

```
================================================================================
CHECKING MAIN PRODUCT FAMILY
================================================================================
main_product_family
Chaudière                          3065
Climatisation                      2438
Poêle                              2027
Pompe à chaleur                    1362
ECS : Chauffe-eau ou adoucisseur    471
Photovoltaïque                      216
Appareil hybride                    122
Autres                               79
```

```
Plomberie Sanitaire                    57
Emetteur de chauffage  ou chappe       47
Produit VMC                            37
Unknown                                16
Fumisterie                              7
Name: count, dtype: int64

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191
```



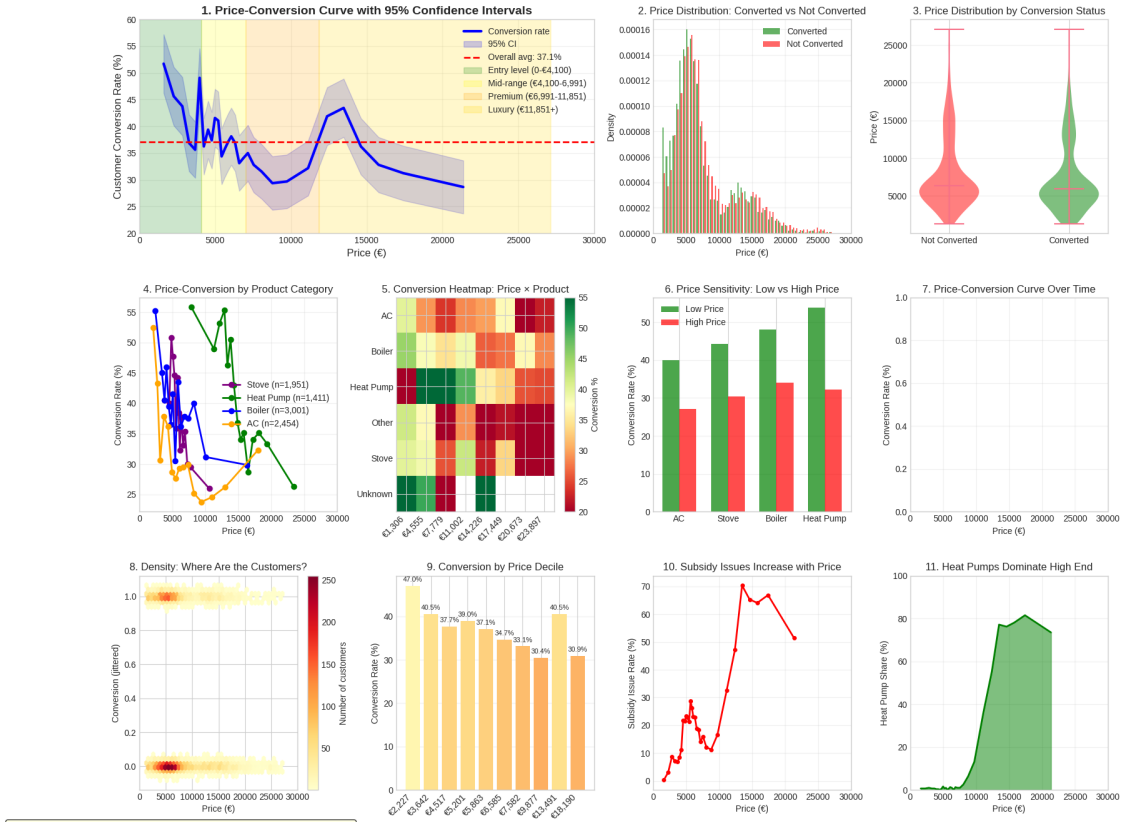

```
================================================================================
Heat Pump Performance Over Time
================================================================================

Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191

Heat pump customers by year:
        conversion_rate  customer_count  avg_price  subsidy_issue_rate
```

```
period
2025                0.406             1646  15068.636                0.755
```



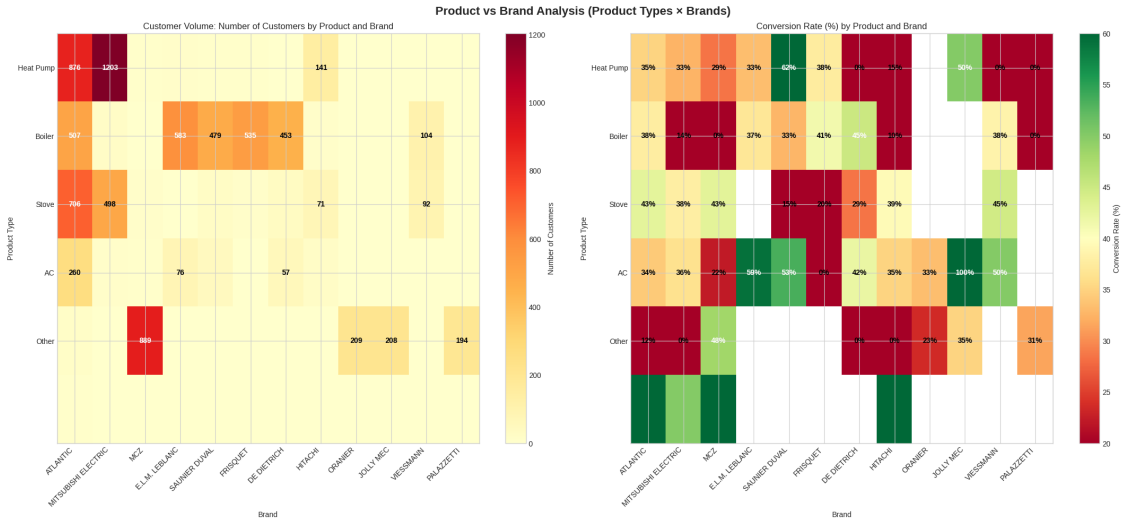

```
Analyzing 9,602 customers
Price range: €1,332 to €27,121
Median price: €6,191

================================================================================
DYNAMIC INSIGHTS CALCULATED:
================================================================================
Sweet spots: €1.6k (51.7%) and €4.0k (49.1%)
Dead zone: €18.5k-27.1k (29-33%)
Heat pumps: 75% subsidy issues, 40.6% conversion
Price sensitivity: Heat Pump (21.5pts) > Boiler (14.0pts) > Stove (13.9pts) > AC
(12.9pts)
Same day decisions: 85%
```

Price-Conversion Dependency Dashboard



Key Insights:
- Two sweet spots: €1.6k (51.7%) and €4.0k (49.1%)
- Dead zone: €18.5k-27.1k (29-33%)
- Heat pumps: 75% subsidy issues, 40.6% conversion
- Price sensitivity: Heat Pump (21.5pts) > Boiler (14.0pts) > Stove (13.9pts) > AC (12.9pts)
- 85% decide in 1 day - price is not the only factor

Product vs Brand Analysis (Product Types × Brands)



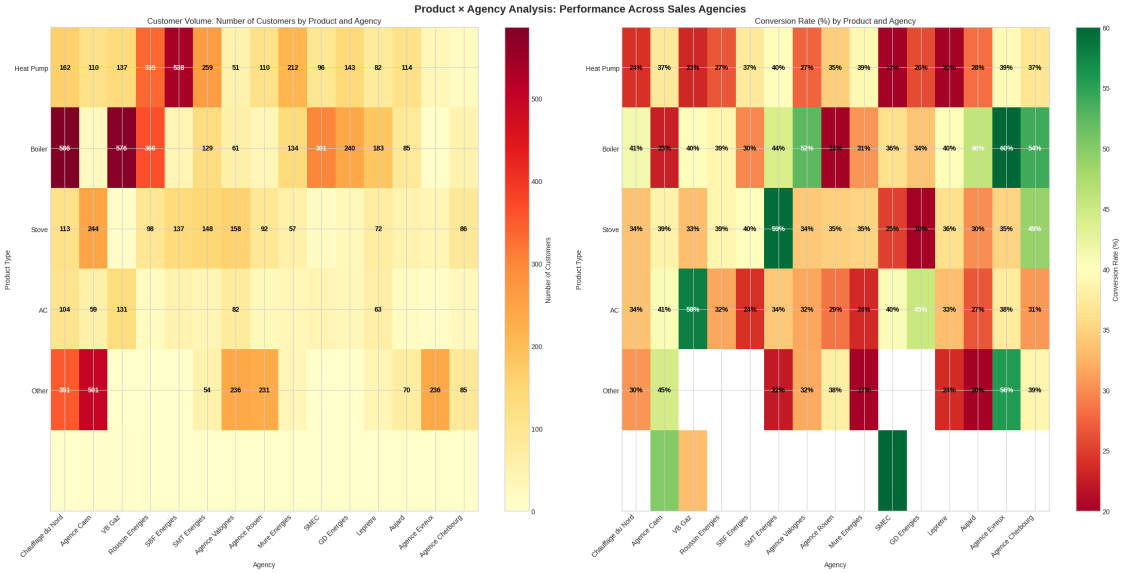
```
================================================================================
PRODUCT × BRAND ANALYSIS
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC  MCZ  E.L.M. LEBLANC
SAUNIER DUVAL  FRISQUET  DE DIETRICH  HITACHI  ORANIER  JOLLY MEC  VIESSMANN
PALAZZETTI
main_equipment_category
AC                            876                 1203    7              12
8         8            2      141        0          2          3           1
Boiler                        507                   21    1             583
479       535          453       10        0          0        104           2
Heat Pump                     706                  498   14               0
27        10           35       71        0          0         92           0
Other                         260                   11    9              76
43         1           57       20        3          1          2           0
Stove                          16                    4  889               0
0         0            1        1      209        208          0         194
Unknown                         9                    2    1               0
0         0            0        1        0          0          0           0

================================================================================
  CONVERSION RATES (%):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC    MCZ  E.L.M. LEBLANC
SAUNIER DUVAL  FRISQUET  DE DIETRICH  HITACHI  ORANIER  JOLLY MEC  VIESSMANN
PALAZZETTI
main_equipment_category
AC                           35.0                 32.5   28.6            33.3
62.5      37.5          0.0     14.9      NaN       50.0        0.0         0.0
Boiler                       37.7                 14.3    0.0            36.9
32.8      41.3         45.0     10.0      NaN        NaN       38.5         0.0
Heat Pump                    42.8                 38.0   42.9             NaN
14.8      20.0         28.6     39.4      NaN        NaN       44.6         NaN
Other                        34.2                 36.4   22.2            59.2
53.5       0.0         42.1     35.0     33.3      100.0       50.0         NaN
Stove                        12.5                  0.0   48.1             NaN
NaN       NaN          0.0      0.0     23.4       35.1        NaN        31.4
Unknown                      66.7                 50.0  100.0             NaN
NaN       NaN          NaN    100.0      NaN        NaN        NaN         NaN

================================================================================
  BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------
```

```
Heat Pump:
  Best brand: VIESSMANN
  Conversion: 44.6%
  Customers: 92
  Top 3 brands:
    1. VIESSMANN: 44.6% (n=92)
    2. MCZ: 42.9% (n=14)
    3. ATLANTIC: 42.8% (n=706)

Boiler:
  Best brand: DE DIETRICH
  Conversion: 45.0%
  Customers: 453
  Top 3 brands:
    1. DE DIETRICH: 45.0% (n=453)
    2. FRISQUET: 41.3% (n=535)
    3. VIESSMANN: 38.5% (n=104)

Stove:
  Best brand: MCZ
  Conversion: 48.1%
  Customers: 889
  Top 3 brands:
    1. MCZ: 48.1% (n=889)
    2. JOLLY MEC: 35.1% (n=208)
    3. PALAZZETTI: 31.4% (n=194)

AC:
  Best brand: SAUNIER DUVAL
  Conversion: 62.5%
  Customers: 8
  Top 3 brands:
    1. SAUNIER DUVAL: 62.5% (n=8)
    2. JOLLY MEC: 50.0% (n=2)
    3. FRISQUET: 37.5% (n=8)

Other:
  Best brand: JOLLY MEC
  Conversion: 100.0%
  Customers: 1
  Top 3 brands:
    1. JOLLY MEC: 100.0% (n=1)
    2. E.L.M. LEBLANC: 59.2% (n=76)
    3. SAUNIER DUVAL: 53.5% (n=43)

================================================================================
 BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------
```

```
ATLANTIC:
  Best product: Unknown
  Conversion: 66.7%
  Customers: 9
  All products:
    Unknown: 66.7% (n=9)
    Heat Pump: 42.8% (n=706)
    Boiler: 37.7% (n=507)
    AC: 35.0% (n=876)
    Other: 34.2% (n=260)
    Stove: 12.5% (n=16)

MITSUBISHI ELECTRIC:
  Best product: Unknown
  Conversion: 50.0%
  Customers: 2
  All products:
    Unknown: 50.0% (n=2)
    Heat Pump: 38.0% (n=498)
    Other: 36.4% (n=11)
    AC: 32.5% (n=1,203)
    Boiler: 14.3% (n=21)
    Stove: 0.0% (n=4)

MCZ:
  Best product: Unknown
  Conversion: 100.0%
  Customers: 1
  All products:
    Unknown: 100.0% (n=1)
    Stove: 48.1% (n=889)
    Heat Pump: 42.9% (n=14)
    AC: 28.6% (n=7)
    Other: 22.2% (n=9)
    Boiler: 0.0% (n=1)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 59.2%
  Customers: 76
  All products:
    Other: 59.2% (n=76)
    Boiler: 36.9% (n=583)
    AC: 33.3% (n=12)

SAUNIER DUVAL:
  Best product: AC
```

```
  Conversion: 62.5%
  Customers: 8
  All products:
    AC: 62.5% (n=8)
    Other: 53.5% (n=43)
    Boiler: 32.8% (n=479)
    Heat Pump: 14.8% (n=27)

FRISQUET:
  Best product: Boiler
  Conversion: 41.3%
  Customers: 535
  All products:
    Boiler: 41.3% (n=535)
    AC: 37.5% (n=8)
    Heat Pump: 20.0% (n=10)
    Other: 0.0% (n=1)

DE DIETRICH:
  Best product: Boiler
  Conversion: 45.0%
  Customers: 453
  All products:
    Boiler: 45.0% (n=453)
    Other: 42.1% (n=57)
    Heat Pump: 28.6% (n=35)
    AC: 0.0% (n=2)
    Stove: 0.0% (n=1)

HITACHI:
  Best product: Unknown
  Conversion: 100.0%
  Customers: 1
  All products:
    Unknown: 100.0% (n=1)
    Heat Pump: 39.4% (n=71)
    Other: 35.0% (n=20)
    AC: 14.9% (n=141)
    Boiler: 10.0% (n=10)
    Stove: 0.0% (n=1)

================================================================================
  STRATEGIC INSIGHTS
================================================================================

Brand Dominance by Product Category:

Heat Pump:
```

```
  Volume leader: ATLANTIC (706 customers, 48.6% share)
  Conversion leader: VIESSMANN (44.6%)

Boiler:
  Volume leader: E.L.M. LEBLANC (583 customers, 21.6% share)
  Conversion leader: DE DIETRICH (45.0%)

Stove:
  Volume leader: MCZ (889 customers, 58.4% share)
  Conversion leader: MCZ (48.1%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (1,203 customers, 53.2% share)
  Conversion leader: SAUNIER DUVAL (62.5%)

Other:
  Volume leader: ATLANTIC (260 customers, 53.8% share)
  Conversion leader: JOLLY MEC (100.0%)

Top 15 agencies by customer volume:
  Chauffage du Nord: 1,316 customers, 35.0% conversion
  Agence Caen: 940 customers, 41.4% conversion
  VB Gaz: 856 customers, 40.2% conversion
  Roussin Energies: 818 customers, 33.6% conversion
  SBF Energies: 762 customers, 36.6% conversion
  SMT Energies: 631 customers, 43.4% conversion
  Agence Valognes: 588 customers, 34.2% conversion
  Agence Rouen: 497 customers, 34.4% conversion
  Mure Energies: 473 customers, 33.4% conversion
  SMEC: 456 customers, 31.4% conversion
  GD Energies: 443 customers, 30.5% conversion
  Lepretre: 438 customers, 33.3% conversion
  Aujard: 334 customers, 31.1% conversion
  Agence Evreux: 319 customers, 51.1% conversion
  Agence Cherbourg: 266 customers, 44.0% conversion
```




```
================================================================================
PRODUCT × AGENCY ANALYSIS
================================================================================

 CUSTOMER VOLUME BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency                Chauffage du Nord  Agence Caen  VB Gaz  Roussin
Energies  SBF Energies  SMT Energies  Agence Valognes  Agence Rouen  Mure
Energies  SMEC  GD Energies  Lepretre  Aujard  Agence Evreux  Agence Cherbourg
main_equipment_category
AC                                   162          110     137
335          538          259                51          110            212
96         143        82     114              33                19
Boiler                               586           22     576
366           37          129                61           36            134
301         240       183      85                5                50
Heat Pump                            113          244       9
98          137          148              158           92             57
16          29        72      50              37                86
Other                                104           59     131
19           50           41                82           28             34
42          31        63      15                8                26
Stove                                351          501       0
0            0           54               236          231             36
0           0        38      70             236                85
Unknown                                0            4       3
```

```
0              0              0                 0              0               0
1             0          0        0               0                  0

================================================================================
  CONVERSION RATES (%) BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency               Chauffage du Nord  Agence Caen  VB Gaz  Roussin
Energies  SBF Energies  SMT Energies  Agence Valognes  Agence Rouen  Mure
Energies   SMEC  GD Energies  Lepretre  Aujard  Agence Evreux  Agence Cherbourg
main_equipment_category
AC                                     24.1         37.3    23.4
26.6          37.4          39.8             27.5          34.5           39.2
12.5         25.9      19.5    28.1           39.4              36.8
Boiler                                 41.1         22.7    40.3
38.8          29.7          44.2             52.5          13.9           30.6
36.2         33.8      40.4    45.9           60.0              54.0
Heat Pump                              33.6         38.5    33.3
38.8          40.1          59.5             34.2          34.8           35.1
25.0         10.3      36.1    30.0           35.1              48.8
Other                                  33.7         40.7    58.0
31.6          24.0          34.1             31.7          28.6           23.5
40.5         45.2      33.3    26.7           37.5              30.8
Stove                                  30.5         44.5     NaN
NaN           NaN          22.2             31.8          38.1           16.7
NaN          NaN      23.7    20.0           55.5              38.8
Unknown                                 NaN         50.0    33.3
NaN           NaN           NaN              NaN           NaN            NaN
100.0          NaN       NaN     NaN            NaN               NaN

================================================================================
  BEST AGENCY BY PRODUCT TYPE
--------------------------------------------------------------------------------
--------------------

Heat Pump:
  Best agency: SMT Energies
  Conversion: 59.5%
  Customers: 148
  Top 3 agencies:
    1. SMT Energies: 59.5% (n=148)
    2. Agence Cherbourg: 48.8% (n=86)
    3. SBF Energies: 40.1% (n=137)

Boiler:
  Best agency: Agence Evreux
  Conversion: 60.0%
  Customers: 5
```

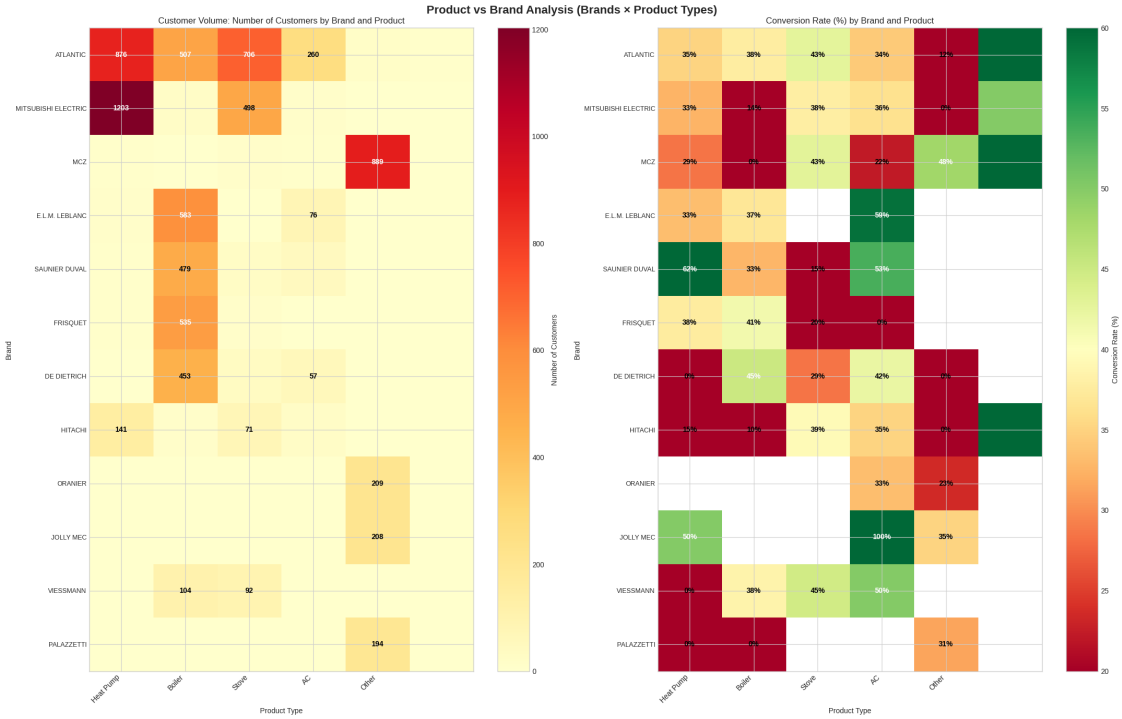
```
  Top 3 agencies:
    1. Agence Evreux: 60.0% (n=5)
    2. Agence Cherbourg: 54.0% (n=50)
    3. Agence Valognes: 52.5% (n=61)

Stove:
  Best agency: Agence Evreux
  Conversion: 55.5%
  Customers: 236
  Top 3 agencies:
    1. Agence Evreux: 55.5% (n=236)
    2. Agence Caen: 44.5% (n=501)
    3. Agence Cherbourg: 38.8% (n=85)

AC:
  Best agency: SMT Energies
  Conversion: 39.8%
  Customers: 259
  Top 3 agencies:
    1. SMT Energies: 39.8% (n=259)
    2. Agence Evreux: 39.4% (n=33)
    3. Mure Energies: 39.2% (n=212)

Other:
  Best agency: VB Gaz
  Conversion: 58.0%
  Customers: 131
  Top 3 agencies:
    1. VB Gaz: 58.0% (n=131)
    2. GD Energies: 45.2% (n=31)
    3. Agence Caen: 40.7% (n=59)

================================================================================
 BEST PRODUCT BY AGENCY
--------------------------------------------------------------------------------
--------------------

Chauffage du Nord:
  Best product: Boiler
  Conversion: 41.1%
  Customers: 586
  All products:
    Boiler: 41.1% (n=586)
    Other: 33.7% (n=104)
    Heat Pump: 33.6% (n=113)
    Stove: 30.5% (n=351)
    AC: 24.1% (n=162)
```

```
Agence Caen:
  Best product: Unknown
  Conversion: 50.0%
  Customers: 4
  All products:
    Unknown: 50.0% (n=4)
    Stove: 44.5% (n=501)
    Other: 40.7% (n=59)
    Heat Pump: 38.5% (n=244)
    AC: 37.3% (n=110)
    Boiler: 22.7% (n=22)

VB Gaz:
  Best product: Other
  Conversion: 58.0%
  Customers: 131
  All products:
    Other: 58.0% (n=131)
    Boiler: 40.3% (n=576)
    Heat Pump: 33.3% (n=9)
    Unknown: 33.3% (n=3)
    AC: 23.4% (n=137)

Roussin Energies:
  Best product: Boiler
  Conversion: 38.8%
  Customers: 366
  All products:
    Boiler: 38.8% (n=366)
    Heat Pump: 38.8% (n=98)
    Other: 31.6% (n=19)
    AC: 26.6% (n=335)

SBF Energies:
  Best product: Heat Pump
  Conversion: 40.1%
  Customers: 137
  All products:
    Heat Pump: 40.1% (n=137)
    AC: 37.4% (n=538)
    Boiler: 29.7% (n=37)
    Other: 24.0% (n=50)

SMT Energies:
  Best product: Heat Pump
  Conversion: 59.5%
  Customers: 148
  All products:
```

```
    Heat Pump: 59.5% (n=148)
    Boiler: 44.2% (n=129)
    AC: 39.8% (n=259)
    Other: 34.1% (n=41)
    Stove: 22.2% (n=54)

Agence Valognes:
  Best product: Boiler
  Conversion: 52.5%
  Customers: 61
  All products:
    Boiler: 52.5% (n=61)
    Heat Pump: 34.2% (n=158)
    Stove: 31.8% (n=236)
    Other: 31.7% (n=82)
    AC: 27.5% (n=51)

Agence Rouen:
  Best product: Stove
  Conversion: 38.1%
  Customers: 231
  All products:
    Stove: 38.1% (n=231)
    Heat Pump: 34.8% (n=92)
    AC: 34.5% (n=110)
    Other: 28.6% (n=28)
    Boiler: 13.9% (n=36)

Mure Energies:
  Best product: AC
  Conversion: 39.2%
  Customers: 212
  All products:
    AC: 39.2% (n=212)
    Heat Pump: 35.1% (n=57)
    Boiler: 30.6% (n=134)
    Other: 23.5% (n=34)
    Stove: 16.7% (n=36)

SMEC:
  Best product: Unknown
  Conversion: 100.0%
  Customers: 1
  All products:
    Unknown: 100.0% (n=1)
    Other: 40.5% (n=42)
    Boiler: 36.2% (n=301)
    Heat Pump: 25.0% (n=16)
```

```
    AC: 12.5% (n=96)

================================================================================
  AGENCY PERFORMANCE SUMMARY
================================================================================

Top 15 agencies by volume:
                    conversion_rate  customer_count  heat_pump_pct
subsidy_issue_pct  avg_price
main_agency
Chauffage du Nord             35.0            1316            9.8
33.6   6361.803
Agence Caen                   41.4             940           28.1
38.3   8796.700
VB Gaz                        40.2             856            1.3
1.3   5009.346
Roussin Energies              33.6             818           14.1
12.2   7878.227
SBF Energies                  36.6             762           22.3
17.2   9083.253
SMT Energies                  43.4             631           28.1
38.0   7026.170
Agence Valognes               34.2             588           32.0
42.3   9495.361
Agence Rouen                  34.4             497           22.1
43.1   8648.829
Mure Energies                 33.4             473           14.4
13.7   6928.316
SMEC                          31.4             456            4.2
9.0   6370.170
GD Energies                   30.5             443            9.7
8.8   5717.904
Lepretre                      33.3             438           20.1
25.3   6999.096
Aujard                        31.1             334           15.9
13.8   7734.909
Agence Evreux                 51.1             319           12.5
41.4   6957.683
Agence Cherbourg              44.0             266           35.7
42.1   8805.974

================================================================================
  STRATEGIC INSIGHTS
================================================================================

Agency Excellence by Product Category:

Heat Pump:
```

```
  Conversion leader: SMT Energies (59.5%, n=148)

Boiler:
  Conversion leader: Agence Evreux (60.0%, n=5)

Stove:
  Conversion leader: Agence Evreux (55.5%, n=236)

AC:
  Conversion leader: SMT Energies (39.8%, n=259)

Other:
  Conversion leader: VB Gaz (58.0%, n=131)

================================================================================
 HEAT PUMP SPECIALISTS
================================================================================

Top 5 agencies for Heat Pump conversion:
  SMT Energies: 59.5% (n=148)
  Agence Cherbourg: 48.8% (n=86)
  SBF Energies: 40.1% (n=137)
  Roussin Energies: 38.8% (n=98)
  Agence Caen: 38.5% (n=244)
```



```
================================================================================
BRAND × PRODUCT ANALYSIS (Swapped Axes)
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
main_brand
ATLANTIC                  876     507        706    260     16        9
MITSUBISHI ELECTRIC      1203      21        498     11      4        2
MCZ                         7       1         14      9    889        1
E.L.M. LEBLANC             12     583          0     76      0        0
SAUNIER DUVAL               8     479         27     43      0        0
FRISQUET                    8     535         10      1      0        0
DE DIETRICH                 2     453         35     57      1        0
HITACHI                   141      10         71     20      1        1
ORANIER                     0       0          0      3    209        0
JOLLY MEC                   2       0          0      1    208        0
VIESSMANN                   3     104         92      2      0        0
PALAZZETTI                  1       2          0      0    194        0

================================================================================
  CONVERSION RATES (%):
--------------------------------------------------------------------------------
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
main_brand
ATLANTIC                 35.0    37.7       42.8   34.2   12.5     66.7
MITSUBISHI ELECTRIC      32.5    14.3       38.0   36.4    0.0     50.0
MCZ                      28.6     0.0       42.9   22.2   48.1    100.0
E.L.M. LEBLANC           33.3    36.9        NaN   59.2    NaN      NaN
SAUNIER DUVAL            62.5    32.8       14.8   53.5    NaN      NaN
FRISQUET                 37.5    41.3       20.0    0.0    NaN      NaN
DE DIETRICH               0.0    45.0       28.6   42.1    0.0      NaN
HITACHI                  14.9    10.0       39.4   35.0    0.0    100.0
ORANIER                   NaN     NaN        NaN   33.3   23.4      NaN
JOLLY MEC                50.0     NaN        NaN  100.0   35.1      NaN
VIESSMANN                 0.0    38.5       44.6   50.0    NaN      NaN
PALAZZETTI                0.0     0.0        NaN    NaN   31.4      NaN

================================================================================
  BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Unknown
  Conversion: 66.7%
  Customers: 9
```

```
  Top 3 products:
    1. Unknown: 66.7% (n=9)
    2. Heat Pump: 42.8% (n=706)
    3. Boiler: 37.7% (n=507)

MITSUBISHI ELECTRIC:
  Best product: Unknown
  Conversion: 50.0%
  Customers: 2
  Top 3 products:
    1. Unknown: 50.0% (n=2)
    2. Heat Pump: 38.0% (n=498)
    3. Other: 36.4% (n=11)

MCZ:
  Best product: Unknown
  Conversion: 100.0%
  Customers: 1
  Top 3 products:
    1. Unknown: 100.0% (n=1)
    2. Stove: 48.1% (n=889)
    3. Heat Pump: 42.9% (n=14)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 59.2%
  Customers: 76
  Top 3 products:
    1. Other: 59.2% (n=76)
    2. Boiler: 36.9% (n=583)
    3. AC: 33.3% (n=12)

SAUNIER DUVAL:
  Best product: AC
  Conversion: 62.5%
  Customers: 8
  Top 3 products:
    1. AC: 62.5% (n=8)
    2. Other: 53.5% (n=43)
    3. Boiler: 32.8% (n=479)

FRISQUET:
  Best product: Boiler
  Conversion: 41.3%
  Customers: 535
  Top 3 products:
    1. Boiler: 41.3% (n=535)
    2. AC: 37.5% (n=8)
```

```
    3. Heat Pump: 20.0% (n=10)

DE DIETRICH:
  Best product: Boiler
  Conversion: 45.0%
  Customers: 453
  Top 3 products:
    1. Boiler: 45.0% (n=453)
    2. Other: 42.1% (n=57)
    3. Heat Pump: 28.6% (n=35)

HITACHI:
  Best product: Unknown
  Conversion: 100.0%
  Customers: 1
  Top 3 products:
    1. Unknown: 100.0% (n=1)
    2. Heat Pump: 39.4% (n=71)
    3. Other: 35.0% (n=20)

ORANIER:
  Best product: Other
  Conversion: 33.3%
  Customers: 3
  Top 3 products:
    1. Other: 33.3% (n=3)
    2. Stove: 23.4% (n=209)

JOLLY MEC:
  Best product: Other
  Conversion: 100.0%
  Customers: 1
  Top 3 products:
    1. Other: 100.0% (n=1)
    2. AC: 50.0% (n=2)
    3. Stove: 35.1% (n=208)

================================================================================
 BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

Heat Pump:
  Best brand: VIESSMANN
  Conversion: 44.6%
  Customers: 92
  Top 5 brands:
    1. VIESSMANN: 44.6% (n=92)
    2. MCZ: 42.9% (n=14)
```

```
    3. ATLANTIC: 42.8% (n=706)
    4. HITACHI: 39.4% (n=71)
    5. MITSUBISHI ELECTRIC: 38.0% (n=498)

Boiler:
  Best brand: DE DIETRICH
  Conversion: 45.0%
  Customers: 453
  Top 5 brands:
    1. DE DIETRICH: 45.0% (n=453)
    2. FRISQUET: 41.3% (n=535)
    3. VIESSMANN: 38.5% (n=104)
    4. ATLANTIC: 37.7% (n=507)
    5. E.L.M. LEBLANC: 36.9% (n=583)

Stove:
  Best brand: MCZ
  Conversion: 48.1%
  Customers: 889
  Top 5 brands:
    1. MCZ: 48.1% (n=889)
    2. JOLLY MEC: 35.1% (n=208)
    3. PALAZZETTI: 31.4% (n=194)
    4. ORANIER: 23.4% (n=209)
    5. ATLANTIC: 12.5% (n=16)

AC:
  Best brand: SAUNIER DUVAL
  Conversion: 62.5%
  Customers: 8
  Top 5 brands:
    1. SAUNIER DUVAL: 62.5% (n=8)
    2. JOLLY MEC: 50.0% (n=2)
    3. FRISQUET: 37.5% (n=8)
    4. ATLANTIC: 35.0% (n=876)
    5. E.L.M. LEBLANC: 33.3% (n=12)

Other:
  Best brand: JOLLY MEC
  Conversion: 100.0%
  Customers: 1
  Top 5 brands:
    1. JOLLY MEC: 100.0% (n=1)
    2. E.L.M. LEBLANC: 59.2% (n=76)
    3. SAUNIER DUVAL: 53.5% (n=43)
    4. VIESSMANN: 50.0% (n=2)
    5. DE DIETRICH: 42.1% (n=57)
```

```
================================================================================
  STRATEGIC INSIGHTS
================================================================================

Brand Performance by Product Category:

ATLANTIC:
  Unknown: 66.7% (n=9)
  Heat Pump: 42.8% (n=706)
  Boiler: 37.7% (n=507)

MITSUBISHI ELECTRIC:
  Unknown: 50.0% (n=2)
  Heat Pump: 38.0% (n=498)
  Other: 36.4% (n=11)

MCZ:
  Unknown: 100.0% (n=1)
  Stove: 48.1% (n=889)
  Heat Pump: 42.9% (n=14)

E.L.M. LEBLANC:
  Other: 59.2% (n=76)
  Boiler: 36.9% (n=583)
  AC: 33.3% (n=12)

SAUNIER DUVAL:
  AC: 62.5% (n=8)
  Other: 53.5% (n=43)
  Boiler: 32.8% (n=479)

FRISQUET:
  Boiler: 41.3% (n=535)
  AC: 37.5% (n=8)
  Heat Pump: 20.0% (n=10)

DE DIETRICH:
  Boiler: 45.0% (n=453)
  Other: 42.1% (n=57)
  Heat Pump: 28.6% (n=35)

HITACHI:
  Unknown: 100.0% (n=1)
  Heat Pump: 39.4% (n=71)
  Other: 35.0% (n=20)

================================================================================
  MARKET LEADERS BY PRODUCT CATEGORY
```

```
================================================================================

Heat Pump:
  Volume leader: ATLANTIC (706 customers, 48.6% share)
  Conversion leader: VIESSMANN (44.6%)

Boiler:
  Volume leader: E.L.M. LEBLANC (583 customers, 21.6% share)
  Conversion leader: DE DIETRICH (45.0%)

Stove:
  Volume leader: MCZ (889 customers, 58.4% share)
  Conversion leader: MCZ (48.1%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (1,203 customers, 53.2% share)
  Conversion leader: SAUNIER DUVAL (62.5%)

Other:
  Volume leader: ATLANTIC (260 customers, 53.8% share)
  Conversion leader: JOLLY MEC (100.0%)

 (main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
  main_brand
  ATLANTIC                  876     507        706    260     16        9
  MITSUBISHI ELECTRIC      1203      21        498     11      4        2
  MCZ                         7       1         14      9    889        1
  E.L.M. LEBLANC             12     583          0     76      0        0
  SAUNIER DUVAL               8     479         27     43      0        0
  FRISQUET                    8     535         10      1      0        0
  DE DIETRICH                 2     453         35     57      1        0
  HITACHI                   141      10         71     20      1        1
  ORANIER                     0       0          0      3    209        0
  JOLLY MEC                   2       0          0      1    208        0
  VIESSMANN                   3     104         92      2      0        0
  PALAZZETTI                  1       2          0      0    194        0,
  main_equipment_category         AC     Boiler  Heat Pump       Other  \
  main_brand
  ATLANTIC                 35.045662  37.672584  42.776204   34.230769
  MITSUBISHI ELECTRIC      32.502078  14.285714  37.951807   36.363636
  MCZ                      28.571429   0.000000  42.857143   22.222222
  E.L.M. LEBLANC           33.333333  36.878216        NaN   59.210526
  SAUNIER DUVAL            62.500000  32.776618  14.814815   53.488372
  FRISQUET                 37.500000  41.308411  20.000000    0.000000
  DE DIETRICH               0.000000  45.033113  28.571429   42.105263
  HITACHI                  14.893617  10.000000  39.436620   35.000000
  ORANIER                        NaN        NaN        NaN   33.333333
```

```
JOLLY MEC                 50.000000        NaN        NaN  100.000000
VIESSMANN                  0.000000  38.461538  44.565217   50.000000
PALAZZETTI                 0.000000   0.000000        NaN         NaN

main_equipment_category      Stove     Unknown
main_brand
ATLANTIC                  12.500000   66.666667
MITSUBISHI ELECTRIC        0.000000   50.000000
MCZ                       48.143982  100.000000
E.L.M. LEBLANC                  NaN         NaN
SAUNIER DUVAL                   NaN         NaN
FRISQUET                        NaN         NaN
DE DIETRICH                0.000000         NaN
HITACHI                    0.000000  100.000000
ORANIER                   23.444976         NaN
JOLLY MEC                 35.096154         NaN
VIESSMANN                       NaN         NaN
PALAZZETTI                31.443299         NaN  )
```

```python
import ipynbname
import subprocess
import os
import sys

def export_notebook_to_pdf(overwrite=True, verbose=True):
    """
    Export current notebook to PDF with overwrite control

    Args:
        overwrite (bool): If True, overwrite existing PDF file
        verbose (bool): If True, print progress messages

    Returns:
        bool: True if successful, False otherwise
    """
    try:
        # Get notebook path
        notebook_path = ipynbname.path()
        notebook_path_str = str(notebook_path)
        pdf_path = notebook_path_str.replace('.ipynb', '.pdf')

        if verbose:
            print(f"  Notebook: {notebook_path.name}")
            print(f"  PDF path: {pdf_path}")

        # Check if PDF exists
        if os.path.exists(pdf_path):
```

```
            if overwrite:
                if verbose:
                    print(f"  Removing existing PDF: {pdf_path}")
                os.remove(pdf_path)
            else:
                if verbose:
                    print(f"  PDF already exists and overwrite=False")
                return False

        # Convert to PDF
        if verbose:
            print("  Converting to PDF...")

        result = subprocess.run([
            'jupyter', 'nbconvert', '--to', 'pdf',
            '--no-prompt',  # Suppress input prompts
            notebook_path_str
        ], capture_output=True, text=True)

        if result.returncode == 0:
            if verbose:
                print(f"  PDF saved: {pdf_path}")
            return True
        else:
            if verbose:
                print(f"  Conversion failed:")
                print(result.stderr)
            return False

    except Exception as e:
        if verbose:
            print(f"  Error: {e}")
        return False

# Use the function
if __name__ == "__main__":
    # Overwrite existing PDF
    export_notebook_to_pdf(overwrite=True)

    # Or create new version without overwriting
    # export_notebook_to_pdf(overwrite=False)export_full_notebook_to_pdf()
```

```
  Notebook: _eda_price_2025.ipynb
  PDF path: /home/valeriya/project/homeserve/ai-france-hvac/_eda_price_2025.pdf
  Removing existing PDF: /home/valeriya/project/homeserve/ai-france-
hvac/_eda_price_2025.pdf
  Converting to PDF…
  PDF saved: /home/valeriya/project/homeserve/ai-france-hvac/_eda_price_2025.pdf
```